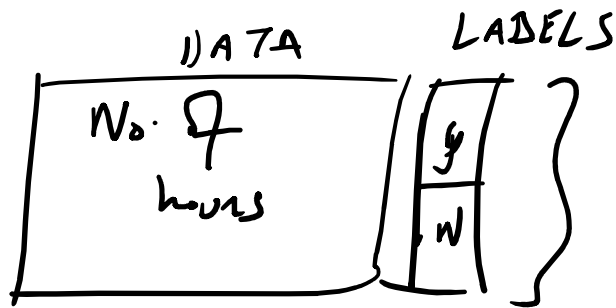


$\Rightarrow$  Classification:

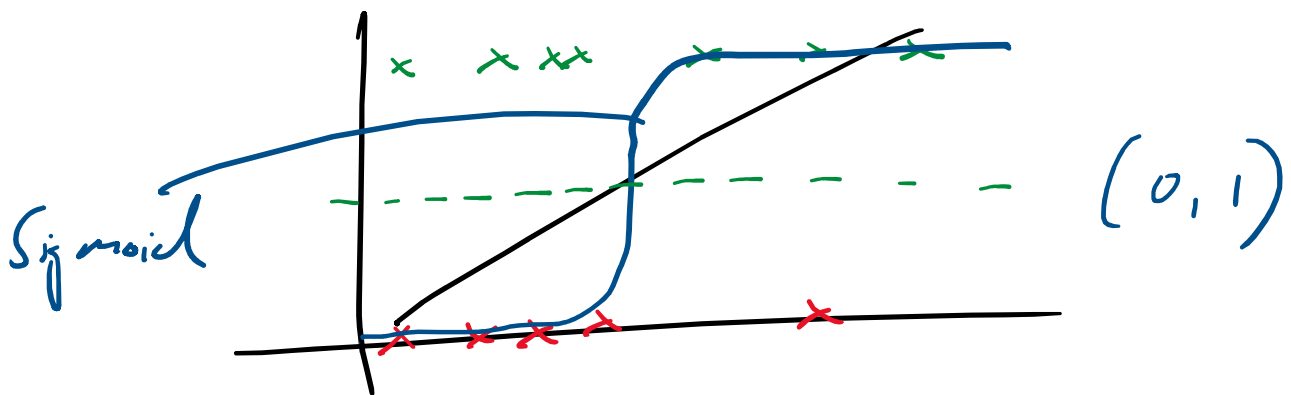
$\Rightarrow$  Binary Classification: 0/1, Yes/No, True/False

$\Rightarrow$  Multiple Classification: N number of classes.



Only discrete values  
(0, 1, 2, 3)  
W.o. decimals

$\Rightarrow$  Logistic Regression:



Convert the linear regression into a logit function.

$$\Rightarrow \text{logit } f(x) = \frac{L}{1 + e^{-k(x-x_0)}}$$

$x_0$  =  $x$  value of midpoint

$L$  = Max Value

$k$  = growth rate

Sigmoid Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \beta_0 + (\beta_1 \cdot x_i)$$

$$\hat{y} = \sigma(z) \rightarrow \text{Yields probability}$$





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→ Example 2:

Hours of Studying (x)      Result (y)

|      |   |
|------|---|
| 0.5  | 0 |
| 1.5  | 0 |
| 2.0  | 0 |
| 4.25 | 1 |
| 3.25 | 1 |
| 5.50 | 1 |

| ⇒ X                              | Y | $(x_i - \bar{x})$ | $(y_i - \bar{y})$ | $(x_i - \bar{x})^2$ | $(y_i - \bar{y})^2$ | $(x_i - \bar{x}) \cdot (y_i - \bar{y})$ |
|----------------------------------|---|-------------------|-------------------|---------------------|---------------------|-----------------------------------------|
| 0.50                             | 0 | -2.38             | -.5               | 5.65                | .25                 | 1.19                                    |
| 1.5                              | 0 | -1.38             | -.5               | 1.9                 | .25                 | 0.69                                    |
| 2.0                              | 0 | -.88              | -.5               | .77                 | .25                 | 0.44                                    |
| 4.25                             | 1 | 1.37              | .5                | 1.87                | .25                 | 0.685                                   |
| 3.25                             | 1 | .37               | .5                | .13                 | .25                 | 0.185                                   |
| 5.50                             | 1 | 2.62              | .5                | 6.86                | .25                 | 1.31                                    |
| $\bar{x} = 2.88$ $\bar{y} = 0.5$ |   |                   |                   | $\Sigma = 17.221$   |                     | $\Sigma = 4.5$                          |

$$\beta_1 = \frac{\sum (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{4.5}{17.221} = \boxed{0.26}$$

$$\beta_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} = \frac{10.26}{17.221} = 0.26$$

$$\beta_0 = \bar{y} - (\beta_1 \cdot \bar{x}) = 0.5 - (0.26 \times 2.88) = -0.25$$

$$z = \beta_0 + (\beta_1 x_i) = -0.25 + (0.26 x_i)$$

$$\hat{y} \approx \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-(-0.25 + (0.26 x_i))}}$$

⇒ Predict the status of examination result where the student was studying for 0.75 hours on daily basis.

1

$$\hat{y} = \frac{1}{1 + e^{-(-0.25 + (0.26 \times 0.75))}}$$

$$= 0.486 \approx 48.6\%$$

⇒ Performance Metric for Classification

Accuracy:

F<sub>1</sub> - Score



Elements of Confusion Matrix.

⇒ Confusion Matrix:

True Positives

False Positives

True Negatives

False Negatives

True Positive

True Negative

|          |                                                                                   |          |
|----------|-----------------------------------------------------------------------------------|----------|
| Actual   | Predicted                                                                         |          |
|          |  |          |
|          | Positive                                                                          | Negative |
| Positive | TP                                                                                | FN       |
| Negative | FP                                                                                | TN       |

$y$

|   |
|---|
| 0 |
| 0 |
| 0 |
| 1 |
| 1 |
| 1 |

$$\hat{y} = \sigma(z)$$

0.4725

0.5377

0.5700

0.2048

0.6477

0.7680

$\hat{y}$

|   |
|---|
| 0 |
| 1 |
| 1 |
| 1 |
| 1 |
| 1 |

|   |           |   |
|---|-----------|---|
|   | $\hat{y}$ |   |
|   | P         | N |
| P | 3         | 0 |
| N | 2         | 1 |

⇒ Precision: Fraction of relevant instances among the retrieved instances.

$$P = \frac{TP}{TP + FP}$$

In our example:

$$P = \frac{TP}{TP + FP} = \frac{3}{3 + 2} = \frac{3}{5} = 0.6 \approx 60\%$$

⇒ Recall: Fraction of relevant instances retrieved.

(Measure of relevant items selected)

$$R = \frac{TP}{TP + FN}$$

$$\boxed{\text{Recall} = \frac{TP}{TP + FN}}$$

$$\frac{TP}{TP + FN} = \frac{3}{3 + 0} = \frac{3}{3} = 1$$

$$\boxed{\approx 100\%}$$

F<sub>1</sub> - Score (Accuracy)

$$\boxed{F_1 = 2 \times \frac{P \times R}{P + R}}$$

$$\text{Eg: } F_1 = 2 \times \frac{0.6 \times 1}{0.6 + 1} = 2 \times \frac{0.6}{1.6}$$

$$= 0.75 \approx \boxed{75\%}$$

⇒ Cost Function (Loss) ~ Classification

→ Cost Function (Loss) ~ Classification

↳ Binary Cross-Entropy (Logarithmic loss)

↳ Categorical Cross-Entropy

log Loss :

$$LL(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N -y_i \cdot \log(P(\hat{y}_i)) - (1-y_i) \cdot \log(1-P(\hat{y}_i))$$

Or example :

y

$\hat{y} = \sigma(z)$

$$-y_i \cdot \log(P(\hat{y}_i)) - (1-y_i) \cdot \log(1-P(\hat{y}_i))$$

0

0.4725

0

0.5377

0

0.5700

1

0.2048

1

0.6477

1

0.7680

$$-0 \cdot \log(0.4725) - (1-0) \cdot \log(1-0.4725)$$

0.6355

$$-0 \cdot \log(0.5377) - (1-0) \cdot \log(1-0.5377)$$

$$= \underline{\underline{0.7716}}$$

$$LL(y, \hat{y}) = \frac{(1.6195 + 0.7716 + 0.8441 + 0.3497 + 0.4743 + 0.2618)}{6}$$

$$= \frac{3.3033}{6} = \boxed{0.5505}$$

$$\approx \boxed{55\%}$$